HUMAN SETTLEMENTS ON PLANETS; NEW STATIONS OR NEW NATIONS

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I. Introduction

In his famous book of 1972, *Manfred Lachs* extensively discussed "the legal situation of objects launched into outer space" and of astronauts[1] and raised the question whether all objects launched into outer space should be placed in a common legal category.[2] He pointed out that there is a difference between the notions of "jurisdiction," on the one hand, and "control," on the other hand,[3] although these two terms are usually quoted together in the space treaties and agreements.

Lachs indicated that space law does not follow the old maritime rule of the "law of the flag" but is based on jurisdiction of the state under whose registry a space object is launched.[4] Already, he foresaw the old conflict between territorial and personal jurisdiction and quoted as an example the "rights of United States nationals in Morocco."[5] He also envisaged emerging problems of space objects permanently established on a celestial body, including the possible changes in ownership which, *inter alia*, would require corresponding changes in the registry with all that this would imply.[6]

If we want to look into future contingencies, we need not be too "science fictioneerish." Plans for a manned mission to Mars within 30 years have already been published. H.H. Koelle, the chairman of the Subcommittee on Lunar Development of the International Academy of Astronautics, recently mailed out a scenario foreseen for the second half of the 21st century which indicates a fully grown lunar settlement with more than 1000 permanent residents.[7]

Before embarking on a discussion of future legal problems, a brief examination of the present legal situation is in order.

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1 MANFRED LACHS, THE LAW OF OUTER SPACE (Sijthoff 1972).

2 *Id.* at 68.

3 *Id.* at 69.

4 *Id.* at 70.

5 *Id.* at 75. (Quoting, 1952 I.C.J. 191).

6 *Id.* at 73.

7 Communication No. 3-93 (July 1993); An earlier version is quoted by Vladimir Kopal, *Some Legal Aspects of the Return to the Moon and Expected Flights to Other Celestial Bodies;* (IAA Paper 0047/1992, Washington, D.C., 1992); *see also "Mars Exploration,"* IAA Symposium at the 44th International Astronautical Congress (Graz, 1993), *Brief News,* 18 J. SPACE L. 188 (1990).

II. Manned Space Stations de Lege Lata

Already the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space including the Moon an other Celestial Bodies of Oct. 10, 1967, known as the Outer Space Treaty, established, among others, the following space legal principles.

a) National appropriation of outer space, including the Moon and other celestial bodies is prohibited, but their exploration and use is permitted;[8]

b) States are responsible for national activities (and activities of their nationals) in outer space[9] and are internationally liable for damage;[10]

c) States "shall retain jurisdiction and control over space objects and over any personnel thereof, while in outer space or on a celestial body" and retain ownership over space objects;[11]

d) A "registration" of space objects is envisioned.

The following space legal agreements repeat these legal principles in a greater detail and a clearer language:

a) the Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space;[12]

b) the Convention on International Liability for Damage Caused by Space Objects;[13]

c) the Convention on the Registration of Space Objects Launched into Outer Space;[14]

d) the Agreement Governing the Activities of States on the Moon and Other Celestial Bodies.[15]

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8 Arts. I and II.

9 *Id.* at art. VI.

10 *Id.* at art. VII.

11 *Id.* at art. VIII.

12 Agreement on the Rescue of Astronauts, the Return of Astronauts, and the Return of Objects Launched into Outer Space, April 22, 1968, 672 U.N.T.S. 119 (hereinafter "Rescue Agreement").

13 Convention on International Liability for Damage Caused by Space Objects, March 29, 1972, 961 U.N.T.S. 187 (hereinafter "Liability Convention").

14 Convention on the Registration of Objects Launched into Outer Space, Jan. 14, 1975, 1023 U.N.T.S. 15 (hereinafter "Registration Convention").

15 Agreement Governing the Activities of Objects Launched into Outer Space, adopted by the U.N. Gen. Assembly on December 5, 1979, opened for signature on Dec. 18, 1979, entered into force July 11, 1984 (not in force for the United States), U.N. Doc. A/RES/34/68 (1979) [hereinafter "Moon Agreement"].

There is, then, no doubt that according to existing space law the following provisions do apply to space stations and their crews:

i) space objects, and especially manned stations established on a celestial body, remain under the ownership, jurisdiction and control of the state of registry;

ii) personnel of a spacecraft remains under the jurisdiction of their own nation and will have to be safely and promptly returned to representatives of the launching authority.

Only the first manned journeys into outer space were conducted within the national activity of one state only, and with personnel of the launching state only. Later on, both USA and former USSR started launching objects with mixed crews into outer space. However, all these space flights only lasted for a limited time, and no station has been established on another celestial body as yet, whether with a crew from one nation only or from two or more nations.

More recently, planned progress in space activities has brought about the notion of "permanently manned civil space stations (for peaceful purposes), on the one hand, and deep space missions - especially to moon and Mars - on the other hand.[16]

For the purpose of regulating problems which would arise or could be foreseen during future international space activities, the Governments of the United States of America, the Member States of the European Space Agency, Japan and Canada signed on Sept. 29, 1988, the "Agreement on Cooperation in the detailed Design, Development, Operation, and Utilization of a permanently manned civil Space Station", the so-called Space Station Freedom Agreement.[17] The following legal principles included therein are of interest:

a) Each partner shall register as space objects the flight elements it provides; and it shall retain jurisdiction and control over those elements.[18]

b) Each partner has the right to provide qualified personnel.[19]

c) There is agreement on a "Cross-Waiver of Liability".[20]

d) Detailed regulations regarding intellectual property are provided.[21]

e) Regarding personnel there are two explicit provisions:

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16 *See* references at *supra* note 7.

17 For a text of the Agreement on Cooperation in the detailed Design, Development, Operation, and Utilization of a permanently manned civil Space Station see U.S. Senate Committee on Commerce, Science and Transportation, SPACE LAW AND RELATED DOCUMENTS 151*ff*. (1990).

18 *Id*. at art. 5.

19 *Id*. at art. 11 (1).

20 *Id*. at art. 16.

21 *Id*. at art. 21.

> i) While all flight partners may exercise criminal jurisdiction over their respective nationals and generally retain jurisdiction over them, then the following sentence is added: "In addition, the United States may exercise criminal jurisdiction over misconduct committed by a non U.S. national in or on a non-U.S. element of the manned base....which endangers the safety of the manned base or the crew members thereon". However, before trial consultations and concurrence of the relevant partner are called for.[22]
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> ii) It is foreseen that a "Code of Conduct for the Space Station Crew, to be developed by all the partners" will be developed.[23] Such a code, however, has not yet been published.

The dissolution of the former USSR brought about the "Minsk Agreement" of December 30, 1991 between the Russian Federation and eight other States of the former Soviet Union. The parties to this agreement pledge to cooperate regarding future (military and civilian) space activities in accordance with international law, with respect to the training of cosmonauts and sharing of costs.They pledge to provide persons, facilities and funds. Special norms of retaining jurisdiction over space flight personnel, however, are not given.[24]

III. The Necessity for New Regulations

Although the space treaties and agreements as quoted above do not deal expressly with "objects....constructed on a celestial body," it has been questioned, whether such objects or installations would remain space objects with all relevant legal implications. Even a spacecraft - surely a space object - after having landed on a celestial body and after disembarkation might no longer be considered a "space object."

Thus*Sephen Gorove* posed the following question:

> First and foremost, the question arises whether an object launched from Earth would lose its legal characterization as a Space Object upon its landing on the Moon, or Mars order another Celestial Body: Would a moon rover or other movable objects, equipment or supplies originating from the Earth cease to be Space Objects and would the relevant space treaty provisions not be applicable to them following such landing? Or would such objects continue to remain space objects and, if so, for how long?

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22 *Id.* at art. 22 (2).

23 *Id.* at art. 11 (2).

24 ECSL Laws, Doc. ISSN 1013-9036 (1993).

He deems it doubtful whether even a movable space object, being made into becoming part of an immovable structure in the form of a station or facility on a Celestial Body would continue to be regarded as a space object.[25]

Thus the exercise of a firm grip by the launching state (state of registry) over durable stations on a celestial body and over the personnel therein might not remain a certainty, and from this, legal consequences might arise.

Especially, the "Draft for a Convention on Manned Space Flight" deals with "Manned Space Flight Operations," including the "embarkation, launch, in orbit, reentry, landing and disembarkation phases." For such space missions a commander shall be appointed to exert sole authority throughout the flight and to use any reasonable and necessary means to achieve the safety and well-being of all persons on board.[26]

Such authority for a space station commander, a proper "code of conduct" once established as well as the multitude of problems that may arise will bring about contingencies not yet experienced and possibly not yet foreseen.

These contingencies will not necessarily arise in all severity regarding space stations in orbit around the Earth or fixed stations on the moon. For such stations could be reached from the surface of the Earth within a few days or even hours. The state of registry would never loose the practical possibility of exerting its jurisdiction and control.

But regarding a possible permanent station on Mars or other planets, the factual situation will be vastly different.

Even under optimal conditions (during the time of a "starting window") a flight to Mars would require months.[27] This same window would not be usable for the return flight. Either a long waiting period or a substantially prolonged time of flight would ensue. All in all, the exercise of actual control and jurisdiction over a Mars station and/or its personnel by a state of registry would require several years.[28]

I do not believe that this scenario could be envisioned as actually having the possibility for an exercise of control.

Therefore, we have to ponder this situation in the light of general international law.

IV. New Code of Conduct - Basis For a New Constitution?

For many years, leading authorities in the field of space law have foreseen problems which would arise in connection with permanently

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25 21 J. SPACE L. 11 (1993).

26 Research project of Böckstiegel, Vereshchetin and Gorove. For a text of the Draft, see 18 J. SPACE L. 209*ff.* (1990).

27 SAENGER, RUMFAHRT HEUTE-MORGEN-ÜBERMORGEN, (Econ Verlag).

28 *Id.* at 266.

manned stations on celestial bodies. We may (in alphabetical order) refer to the papers of Christol,[29] Gorove,[30] Kopal,[31] Matte,[32] Sterns and Tennen,[33] Vereshchetin,[34] and Vitt.[35]

Regarding future permanently manned space objects, there are, as we have seen, two very different categories:

a) Objects near Earth (such as Space Station Freedom, and even a permanently manned lunar station) will carry personnel which may return (or be returned) back to the home planet within a few days, if necessary. Even if an installation on the moon is populated by hundreds of people, there is no factual severance necessary from Earth and its authorities.

b) Inhabitants of manned stations on Mars or its moons or on an asteroid[36] or on any other celestial body within our solar system will have to travel there, stay there and remain off the Earth for several years, at a minimum.

If we establish such a station on Mars, crews of different nations, and of both sexes, will be on board. Married couples will live there and bear children; but married couples may divorce, and new liaisons (marriages or others) might be formed.

The "code of conduct" will then not be restricted to questions of criminal laws and property laws, but "ultimate principles of social orders in space communities" will evolve.

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29 Carl A. Christol, *Space Stations: Present and Future*, 17 PROC. COLLOQ. L. OUTER SPACE 364 (1975).

30 Stephen Gorove, *Legal Aspects of Stations in Space, in* SPACE STATIONS - LEGAL ASPECTS OF SCIENTIFIC AND COMMERCIAL USE IN A FRAMEWORK OF TRANSATLANTIC COOPERATION 143*ff.* (K.-H. Böckstiegel ed. 1985); STEPHEN GOROVE, STUDIES IN SPACE LAW - ITS CHALLENGES AND PROSPECTS 211*ff;* (Sijthoff 1977). *See also* Katherine Gorove, *The U.S./International Space Station Agreement of Sept. 29, 1988: Some Legal Highlights* 15 J. SPACE L.182*ff.* (1988).

31 Vladimir Kopal, *Some Legal Aspects of the Return to the Moon and Expected Flights to other Celestial Bodies* (IAA Paper 92-0047, Washington, D.C. 1992).

32 Nicholas M. Matte, *Space Stations: A Peaceful Use for Humanity?,* 10 ANNALS AIR & SPACE L. 434*ff.* (1985).

33 P.M. Sterns & L.I. Tennen, *International Recognition of the "Art of Living in Spac": The Emergence of Settlement Competence,* 22 PROC. COLLOQ. L OUTER SPACE 221 (1979). P. M. Sterns & L.I. Tennen, *Jurisprudential Philosophies of the Art of Living in Space, the Transnational Imperative* 25 PROC. COLLOQ. L OUTER SPACE 187*ff.* (1982).

34 Vladlen Vereshchetin, *Mir: A Soviet Space Station, Some Legal Aspects of International Cooperation,* PROC. COLLOQ. L. OUTER SPACE 53*ff.* (1988).

35 Von Elmar Vitt, *Die Weltraumkolonie, in* HANDBUCH DES WELTRAUMRECHTS 579*ff.* (Karl-Heinz Böckstiegel ed., Carl Heymanns Verlag 1991).

36 Ernst Fasan, *Large Space Structures and Celestial Bodies,* 27 PROC. COLLOQ. L OUTER SPACE 243*ff.* (1985)..

Personnel of a Mars station will become inhabitants of such a station. The exercise of jurisdiction and control by the state (or states) of registry may become difficult, it may even become impractical and impossible. States may realize that. It may be impractical, to sacrifice billions of dollars and years of time simply to send a spaceship to Mars in order - for instance - to bring a criminal home for trial or to solve civil disputes, or to replace an incapacitated (or undesired) station commander, etc.

On the other hand, new bonds of common interests will necessarily develop between crew members of different nations. Children may be born from international marriages who will have to be raised and educated, and it might prove difficult to establish their nationality in relation to the relevant states of registry.

Conflicts of interests between the states of registry on Earth and the common interests of a crew on a distant space station may develop. Such conflicts may increase, if and when such a station would become autarchic and self sufficient.

It may not be necessary, to repeat the mistakes that led to former frictions between colonies and their homelands. At the same time, one should not foresee some "wars of independence." The vast distances from home would render such wars absolutely senseless. The development of international law and the continuing awareness of being humans from the same home planet would and should prevent this.

There still would remain strong ties with the home countries. Scientific exploration on the planet, on the one hand, and exploitation of possible resources of this planet, on the other, would - irrespective of the vast distances - prove to be a strong connection between the related planet and Earth.

However, with the passage of time, more and more the space station - a true human settlement - would create its own rules.

Once the manned space station has become a lasting settlement on a planet and a new generation grows there and once a new feeling of solidarity with other crew members surpasses the emotional links to the home state, the question may arise if and when such a community might become a new nation in itself.

There is no question that a recognition by other nations - even an acceptance in the community of nations - for instance the U.N., would be proof that a new nation did arise. But according to many theories, recognition by other states is only of a decleratory nature.[37] International law has created certain conditions for a new "State" coming into existence.[38]

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37 A. VERDROSS, VÖLKERRECHT 183*ff.* (Springer 1959).

38 *Id.* at 1980*ff*; SCHLOCHAUSER, WÖRTERBUCH DES VÖLKERRECHTS, 44*ff.* (W. de Gruyter 1960).

Is has been said that the "inner sovereignty" is "prior to and independent of the Treaty."[39]

A new state must necessarily have its own territory, it must have its own people, and it must have its own national organization which is adhered to by the people (notion of effectivity).[40] A state (even prior to recognition) must have a kind of organization, which enables it to live according to international law. If then recognition by other states follows, this would be the declaratory consequence.

There does exist an interrelation between the act of recognition, on the one hand, and the starting of international diplomatic relations, on the other hand.[41]

Recognition in itself is a unilateral act of another nation (or more of them) and this may or may not be a part of a new international treaty between the recognizing nation and the (newly) recognized state.

But the notion of effectivity may give independence to a new state even without (or before) the occurrence of first acts of recognition by third parties.

Regarding the new national territory, difficulties may arise with regards to the non - appropriability of celestial bodies, as discussed above.[42]

But once more here, facts will have normative power. An area on the celestial body will be inhabited; its minerals will be used; inhabitants will move through such territory, and the "rules of conduct" will be valid within such an area.

There is of course, no example in human history applicable to our problem. However, the legal history of the Union of South Africa and, especially, of the Free State of Oranje and of Transvaal demonstrate how a *defacto* development may lead to an independent nation.

The appearance of the Lowlanders around Cape Town in the 17th Century resulted in a fixed settlement, extending its territory more and more toward the East and the North. Even after Great Britain's occupation of Cape Province, and even after the Treaty of August 13, 1814 between the Netherlands and Great Britain, settlers developed regions which later became the territories of the Free State of Oranje and of Transvaal, as mentioned above. Both territories constituted themselves as independent republics and only during the Boore War (1899 - 1902) did they lose their independence.[43]

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39 A. VERDROSS, *supra* note 37, at 182.

40 SCHLOCHAUSER *supra* note 38, at 48.

41 A. VERDROSS, *supra* note 37, SCHLOCHAUSER, *supra* note 38.

42 *See* art. II of the Outer Space Treaty. The notions of "province of all mankind," "common heritage of mankind," and "international regime" seem to point into another direction. But exploration, use and exploitation of planetary resources will nevertheless bring about factual situations similar to an exclusive administration of certain planetary areas.

43 SCHLOCHAUSER, *supra* note 38, at 408.

This example illustrates the coming into existence of these new nations in uncivilized territories. When the Boores moved across the Vaal River, they surely had no fixed plans to become a new nation. Things developed gradually. The same may be true for a settlement, for instance, on Mars.

The severance of such a human settlement on a planet from its home nations (states of registry) by virtue of the enormous distances, the increasing coherence of its members both socially and economically, and the common defense against a hostile environment may build the basis for the emergence of a new organized community.

This may lead to the feeling of "inner sovereignty" among a clearly defined group of people, namely, the inhabitants of the said Mars station. The group of people will have a kind of administrative organization - be it under a "Station Commander" or under any other kind of administrative guidance.

There will have to be some rules, surpassing the "Code of Conduct", and these may become the basis of a new constitution and a new legal system.

Not so easy is the question of national territory to be solved. For as we have learned, celestial bodies are not subject to national appropriation. However, within the borders of the station itself, within the limits of the main activity of this station's personnel in the surrounding areas, a kind of clearly permitted use of a land area will exist.[44] This use will be an exclusive one, because there will not be any competitions, unless we imagine more than one such station in the same stage of development. "Territory" (or rather "areatory") will be available in vast quantities. Therefore, this question of national territory will not be of extreme importance.

If then the inhabitants of such a station decide to act as a community, even toward the old states of registry and if, on the other hand, those states of registry have loosened their influence due to the vast distances, the main conditions for the emergence of a new nation may be there. As a result, and very likely, a kind of "Declaration of Independence" might follow. The new station will thus become a New Nation.

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44 Art. I of the Outer Space Treaty; Art. 2 of the Moon Agreement.